

Aaron Kiem

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Portfolio

SUMMARY

Electrical Systems Engineering senior with hands-on project experience in circuit design, PCB layout, and power electronics simulation. Seeking full-time and internship positions in manufacturing, electrical integration, and other engineering industries supporting HV/LV, UPS, and FMCS.

EDUCATION

Arizona State University, Ira A. Fulton School of Engineering

Tempe, AZ

B.S.E. in Electrical Systems Engineering; GPA: 3.75

Expected May 2027

- Relevant Coursework: Electrical Fundamentals, Design of Electrical Systems, Computational Modeling of Engineering Systems, CADD and Solid Modeling

LANGUAGES & TECHNOLOGIES

Design & Simulation: SOLIDWORKS, AutoCAD, KiCAD, LTSpice, MATLAB, Simulink, Simscape

Electrical Systems: Circuit/schematic design, PCB layout, power electronics simulation, multimeter diagnostics

Hardware Integration: Arduino IDE, MPLAB X IDE, ESP32 and PIC18 microcontrollers

Certifications: Power Electronics Simulation (MathWorks), Circuit Simulation (MathWorks), MATLAB Onramp, Simulink Onramp, Simscape Onramp

PROJECTS

PCB Metal Detection System | Electrical Design Coordinator, 6-person team

Spring 2026

- Designed the power and sensor PCB in KiCAD, integrating ESP32 and PIC18 microcontrollers for metal detection
- Modeled 3D printed mounts in SOLIDWORKS and documented design decisions in a GitHub report
- Managed the team schedule and budget, coordinating deliverables across 6 engineers

Clap-Activated Lighting Control System | Circuit Project Manager, 4-person team

Fall 2025

- Built a 4-subsystem sensor/actuator network with a shared connection hub, successfully integrating all subsystems into one control system
- Driving sound-triggered, touchless, and daylight-based lighting responses in an LED by designing and programming a PIC18F47K42 microcontroller circuit

Five-Box Coded Lock System | Team Lead, 4-person team

Spring 2025

- Led a 4-person team to design and 3D print 5 independently keyed lockboxes controlled from a single Arduino UNO and shared keypad
- Engineered motor-driven locking mechanisms for 5 independent units routed through one central microcontroller circuit

EXPERIENCE

Shift Manager

Sep 2022 – Present

Pizza Hut

Queen Creek, AZ

- Manage opening/closing operations, cash handling, and inventory across 5 restaurant locations while maintaining food-safety compliance (ServSafe certified)

Delivery Driver

May 2025 – Present

DoorDash

Queen Creek – Tempe, AZ

- Independently manage delivery logistics and time-sensitive routing across the Queen Creek–Tempe corridor

ACTIVITIES

Blender & 3D Modeling: Independent projects in 3D modeling, rigging, procedural node texturing, VFX tracking, and LiDAR scanning

3D Printing: Personal print practice using SOLIDWORKS and Sketchfab designs in Ultimaker Cura

Game Design: Independent projects applying AI-assisted coding and troubleshooting